

System Trimming Report: two_link_robotic_arm

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System Information

Number of States	4
Number of Inputs	2
State Variables	theta1, theta2, theta1_dot, theta2_dot
Input Variables	tau1, tau2
Operating Conditions	{'theta1': 0.0}
Parameters	{'m1': 1.0, 'm2': 0.8, 'L1': 0.5, 'L2': 0.4, 'Lc1': 0.25, 'Lc2': 0.2}

Equilibrium (Trim Point)

State Equilibrium (x_e):

theta1	0.000000
theta2	0.000000
theta1_dot	0.000000
theta2_dot	0.000000

Input Equilibrium (u_e):

tau1	0.000000
tau2	0.000000

Linearized Model

A matrix (State Matrix)

0.0000	0.0000	1.0000	0.0000
0.0000	0.0000	0.0000	1.0000
-26.0284	22.9768	-0.7685	2.2324
38.2348	-104.1134	2.2324	-8.8655

B matrix (Input Matrix)

0.0000	0.0000
0.0000	0.0000
7.6853	-22.3239
-22.3239	88.6551

C matrix (Output Matrix)

1.0000	0.0000	0.0000	0.0000
0.0000	1.0000	0.0000	0.0000
0.0000	0.0000	1.0000	0.0000
0.0000	0.0000	0.0000	1.0000

D matrix (Feedthrough Matrix)

0.0000	0.0000
0.0000	0.0000
0.0000	0.0000
0.0000	0.0000

Stability Analysis

Classification: asymptotically stable

Poles / Eigenvalues:

Pole	Value	$ \lambda $
Pole 1	-4.7180 + 9.5804j	10.6791
Pole 2	-4.717967 + -9.580422j	N/A
Pole 3	-0.0990 + 4.0061j	4.0073
Pole 4	-0.099050 + -4.006104j	N/A

Diagnostics

Converged	Yes
Feasible	Yes
Timestamp	2026-02-09T09:52:14.932757